2026 Creative Engineering Design

ROBOCON

The Odyssey

Guidebook

(Version 2026.11.12)

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ROBOCON Intro.

You are Odysseus travelling to Ithaca!

- Your ship will travel through Mediterranean Sea where you will lose crews on the way.
- When your ship stops at each trial, some of the crews will be dragged to the death land of Hades.
- Once all trials are completed, move to Ithaca and save Penelope







Game Contents

ROBOCON Playing Field

Monitoring Area

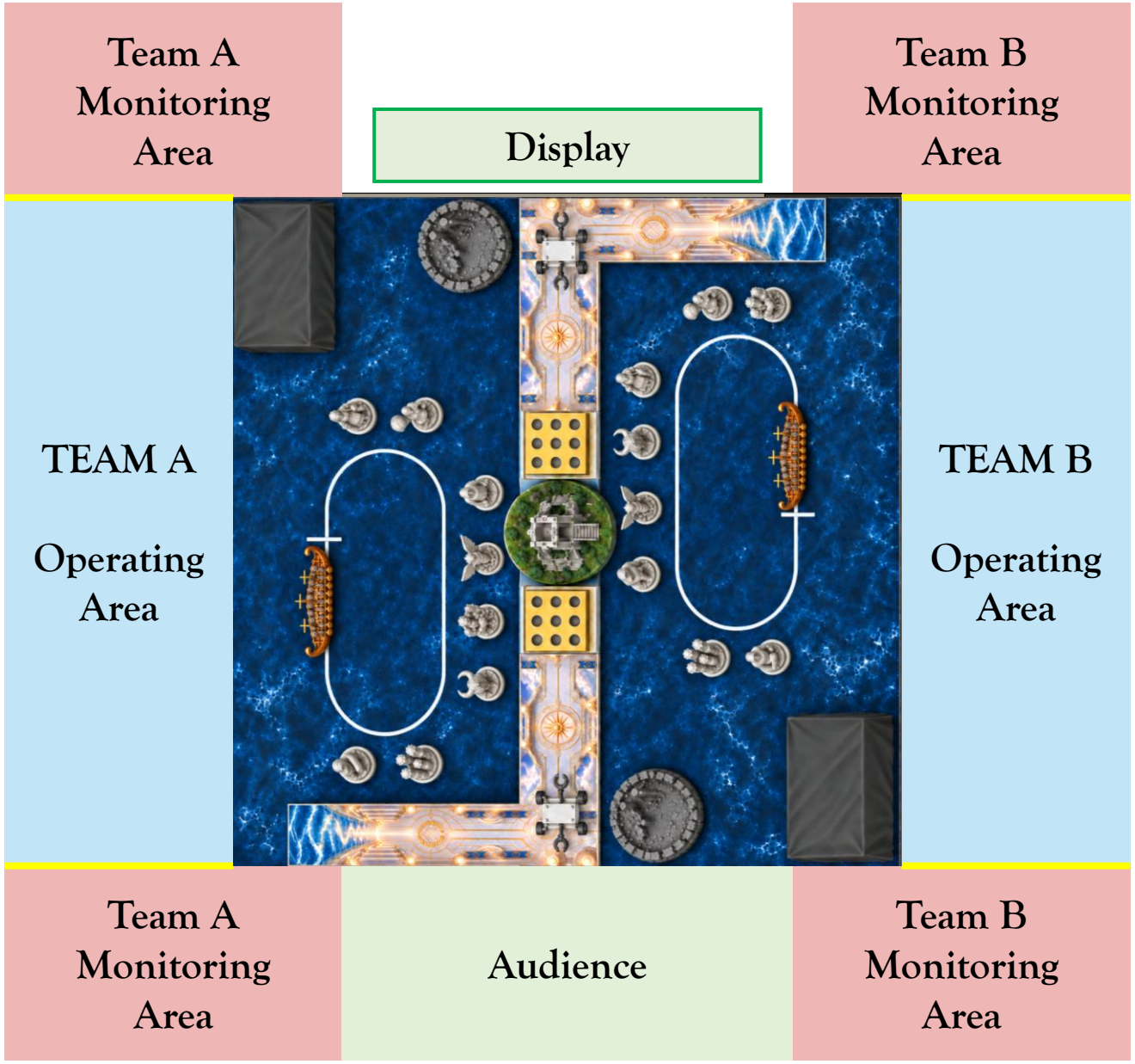
- Only ONE member per team can be inside the monitoring area
- The member can freely move around inside the monitoring area but should never cover the screen

Operating Area

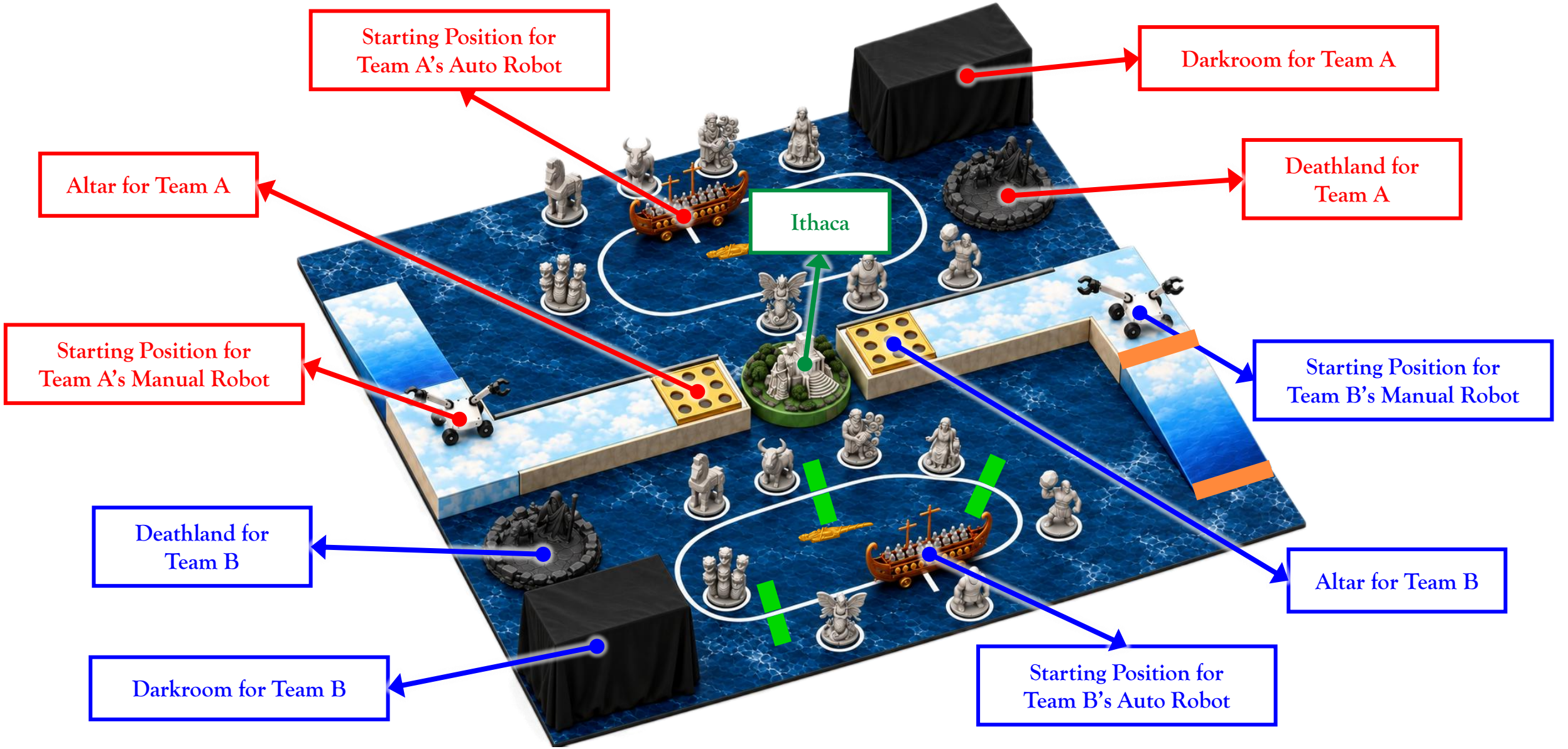
- All control operations [Teleoperations, RC Control] must be conducted within the operating area
- Each Team will be given movable standing desk which can be freely moved within the operating area

Playing Field

- Students cannot enter the playing field during the gameplay
- Students are ONLY allowed to enter at the beginning and the end of the gameplay
- Detail is covered in the next page.



ROBOCON Playing Field



(Size, weight, and material details are in 'Supplementary Information')

Auto & Manual Robots

Auto Robot



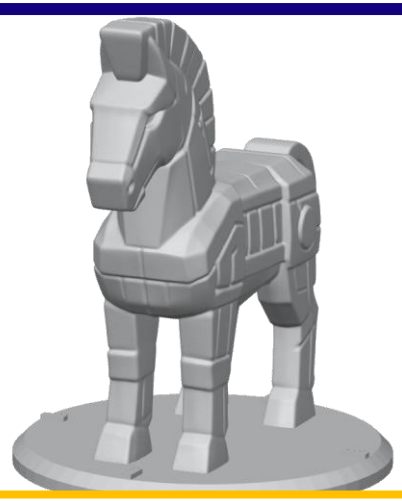
- The Auto Robot must have space for crews on board
- The Auto Robot must be equipped with Jetson Orin Nano with Camera
- It needs to fully autonomous
- **FIND** the next Trial and wait until the trial is completed by Manual Robot

Manual Robot



- The Manual Robot completes trial by removing crews from the ship and place them in the deathland of Hades
- If crews fell in the sea, Manual Robot can rescue them and move them back to the ship
- Once all trials are completed, Manual Robot will help Odysseus to go back to Ithaca and save Penelope

Statues for Trials



Trojan Horse



Cattle of Helios



Calypso



Scylla



Poseidon's Wrath



Sirens



Cyclops



Laestrygonians

Other Important Objects



Soldiers



Odysseus



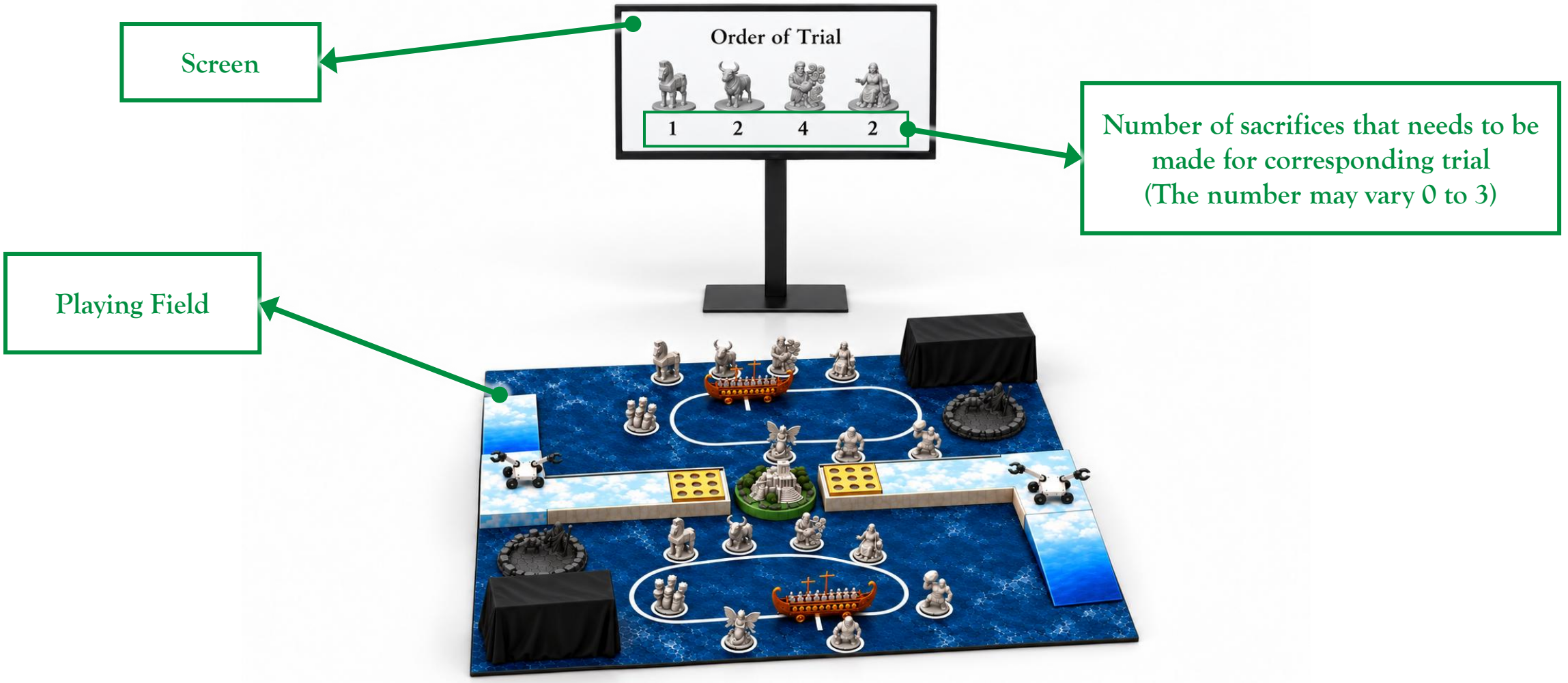
Athena Pin



The Bow

Game Flow

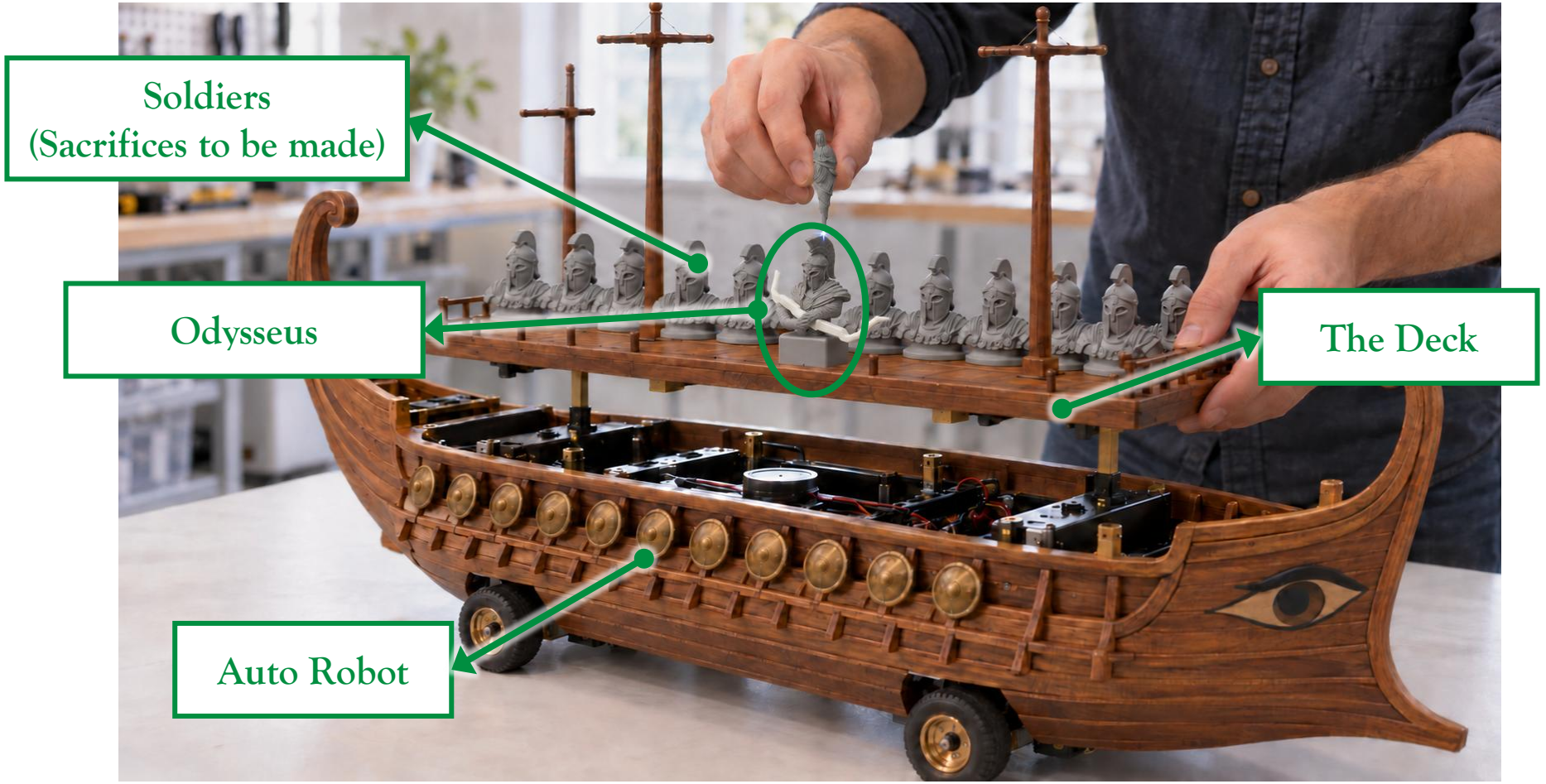
1. At the start signal, the **order of trial** and corresponding **number of sacrifices** will be displayed on the screen. Students will be given **1 minute** to update their code accordingly.
   *Note that the total number of trials may vary from 3 to 8 randomly



Game Flow

2. TAs will load the Auto Robot with the deck with Odysseus and soldiers on board. The students are required to place action camera on their manual robot

*Camera must be mounted on the robot without damaging (using glues or adhesives directly) on camera



Game Flow

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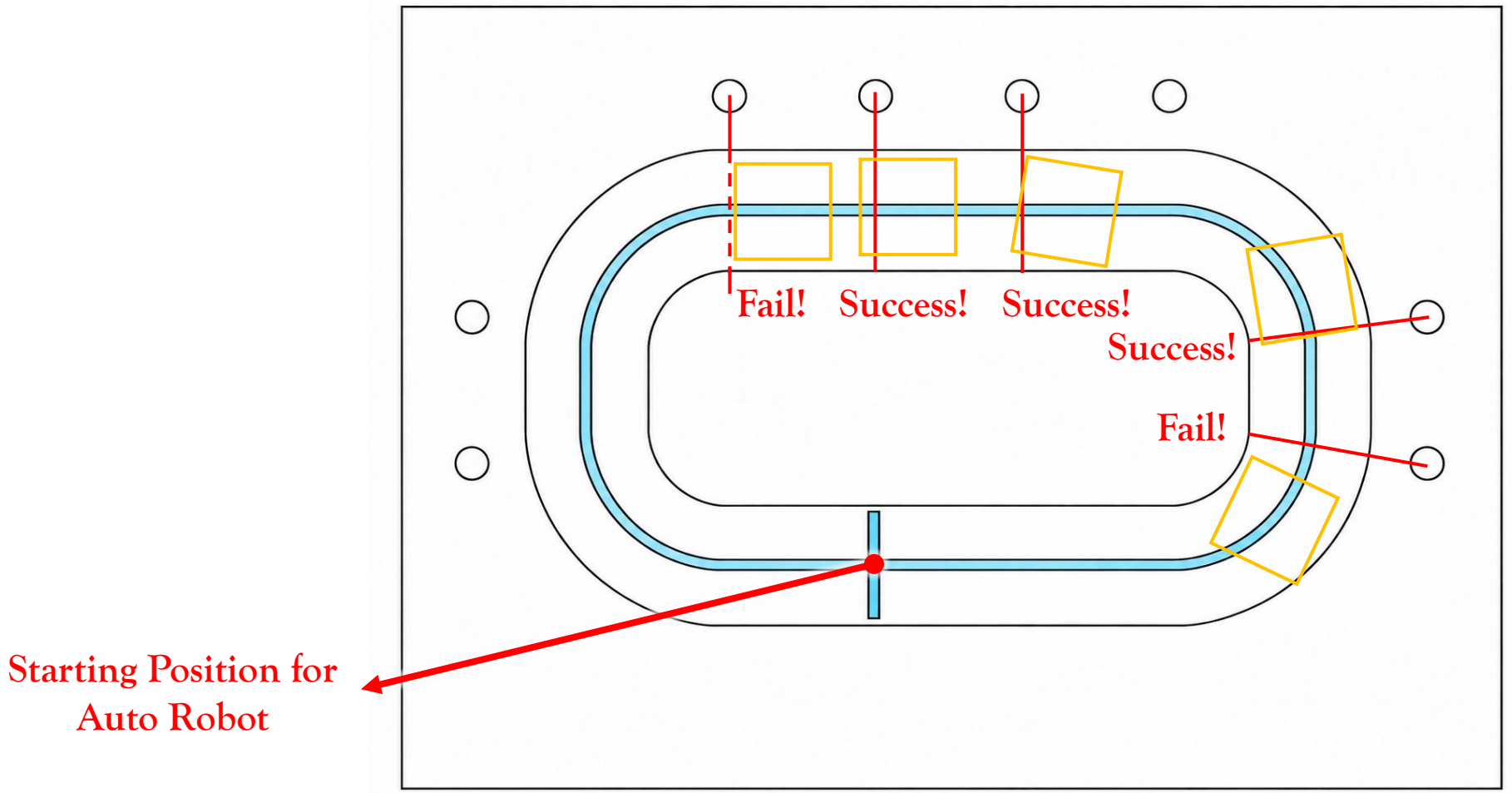
Game Flow

3. The Auto Robot starts moving and **STOPS** when it sees the statue of interest.
   *When the full track is covered by Auto Robot, extra points will be given to the team



Game Flow

3. The Auto Robot starts moving and **STOPS** when it sees the statue of interest.
   *There will be a line connecting the statue and the line track (normal to the line) and if and only if part of the robot is stepping on the mark, it will be considered successful stop. When the full track is covered by Auto Robot, extra points will be given to the team



Game Flow

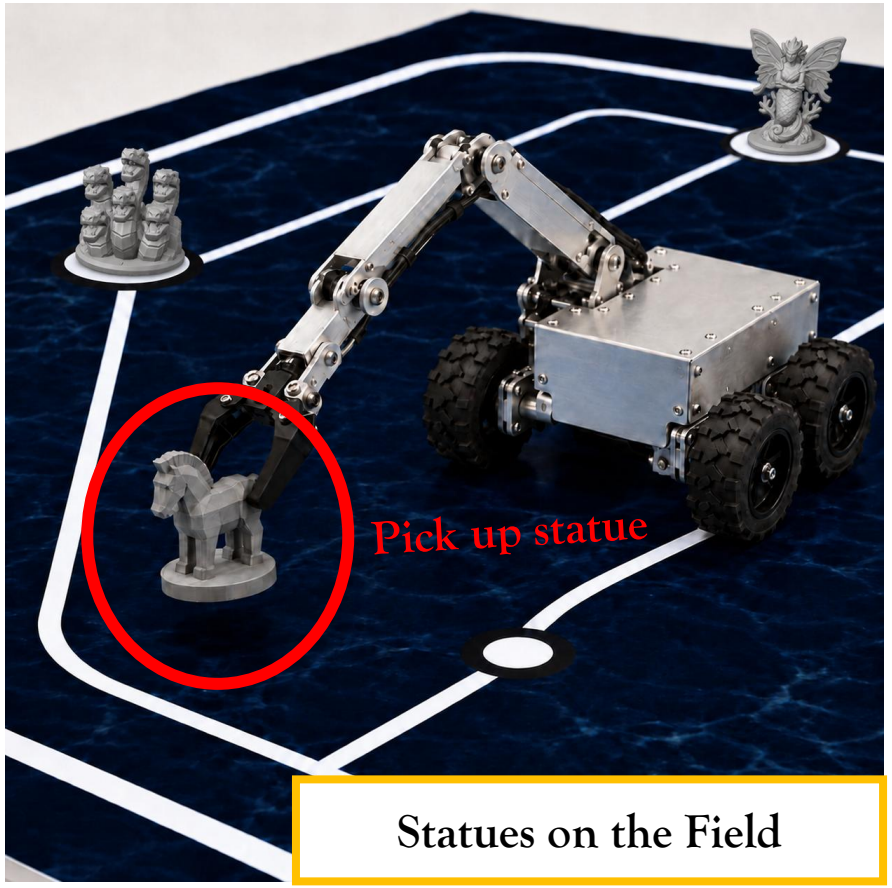
4. When Auto Robot stops at the statue, the Manual Robot will approach Auto Robot and transfer soldiers from Auto Robot to Deathland of Hades





Game Flow

5. After successfully moving required number of soldiers to the Deathland of Hades for each trial, the Manual Robot will transfer **trial statue** to the **altar in Olympus** to indicate completion of trial



Statues on the Field



Altar in Upper Field

Game Flow

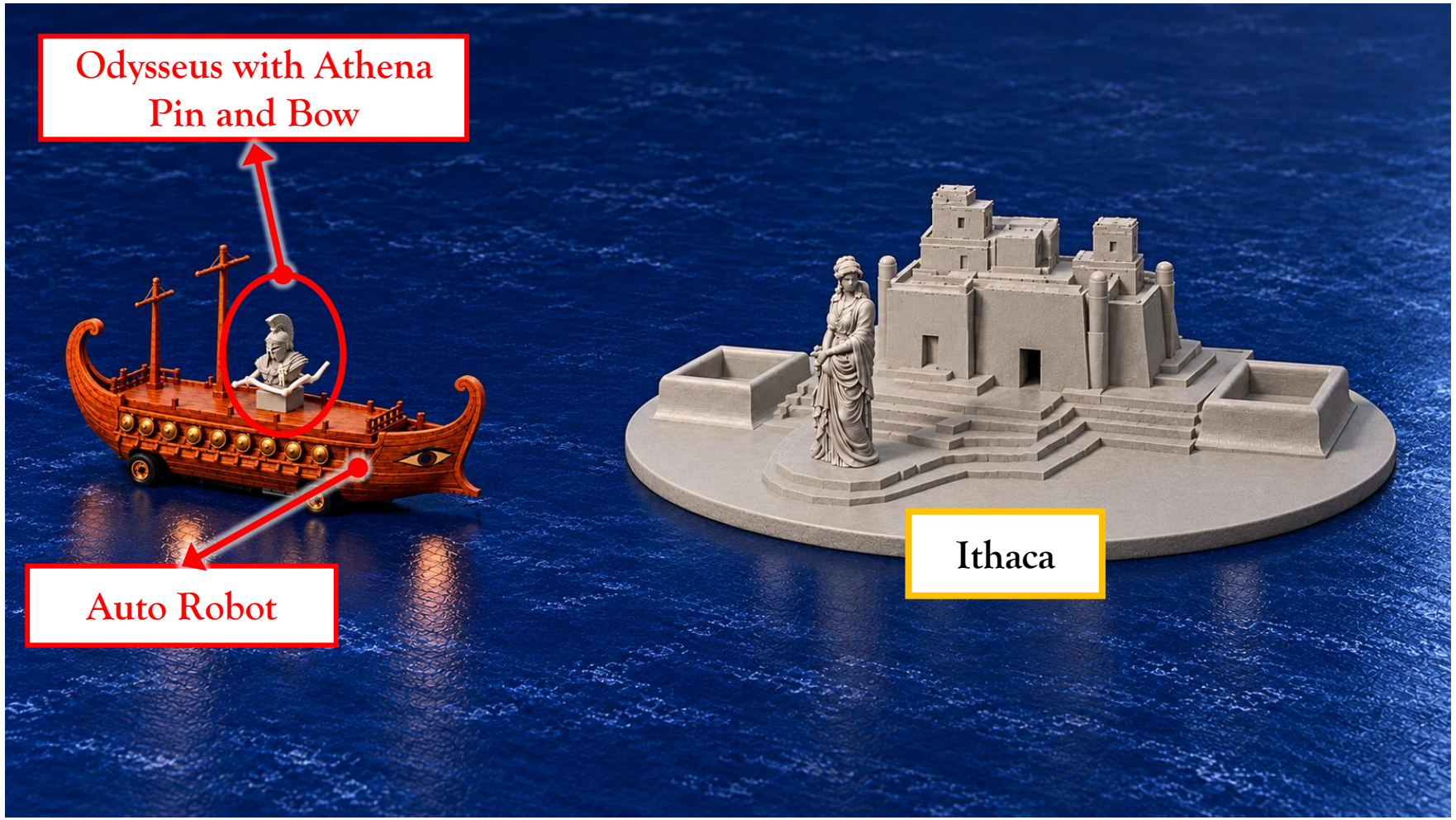
6. After repeating step 3 to 5, all trials are completed (Altar is filled up with statues)
   *Students can place statues anywhere in the altar, there is no preset for the location



Fully filled up Altar in the Upper Field

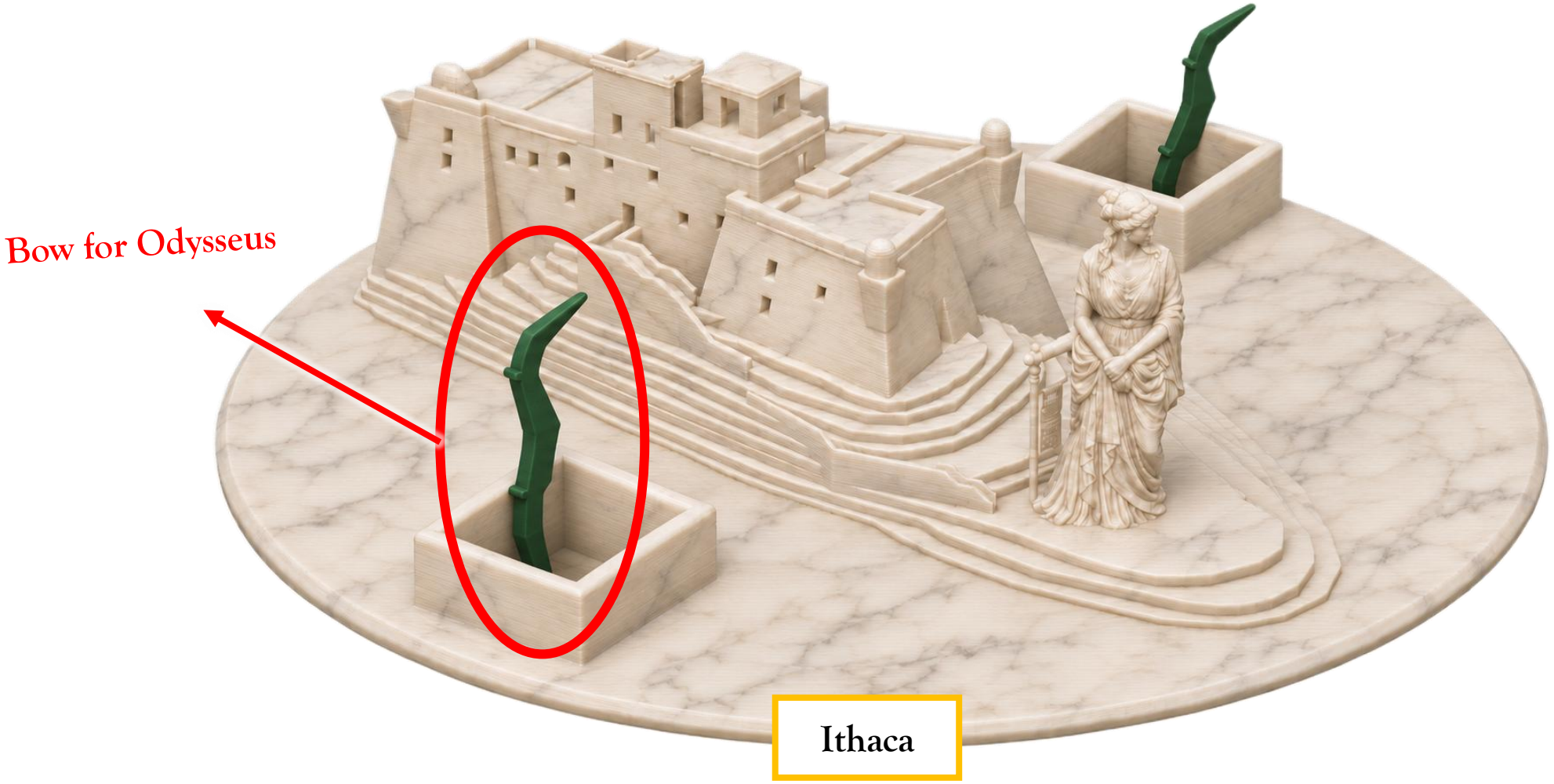
Game Flow

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   *Students can place statues anywhere in the altar, there is no preset for the location



Game Flow

7. The Manual Robot will pick up a bow from Ithaca and transfer to Odysseus prior to entering
   *The bow is placed in the Ithaca where Odysseus must enter as shown below



Game Flow

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Game Flow

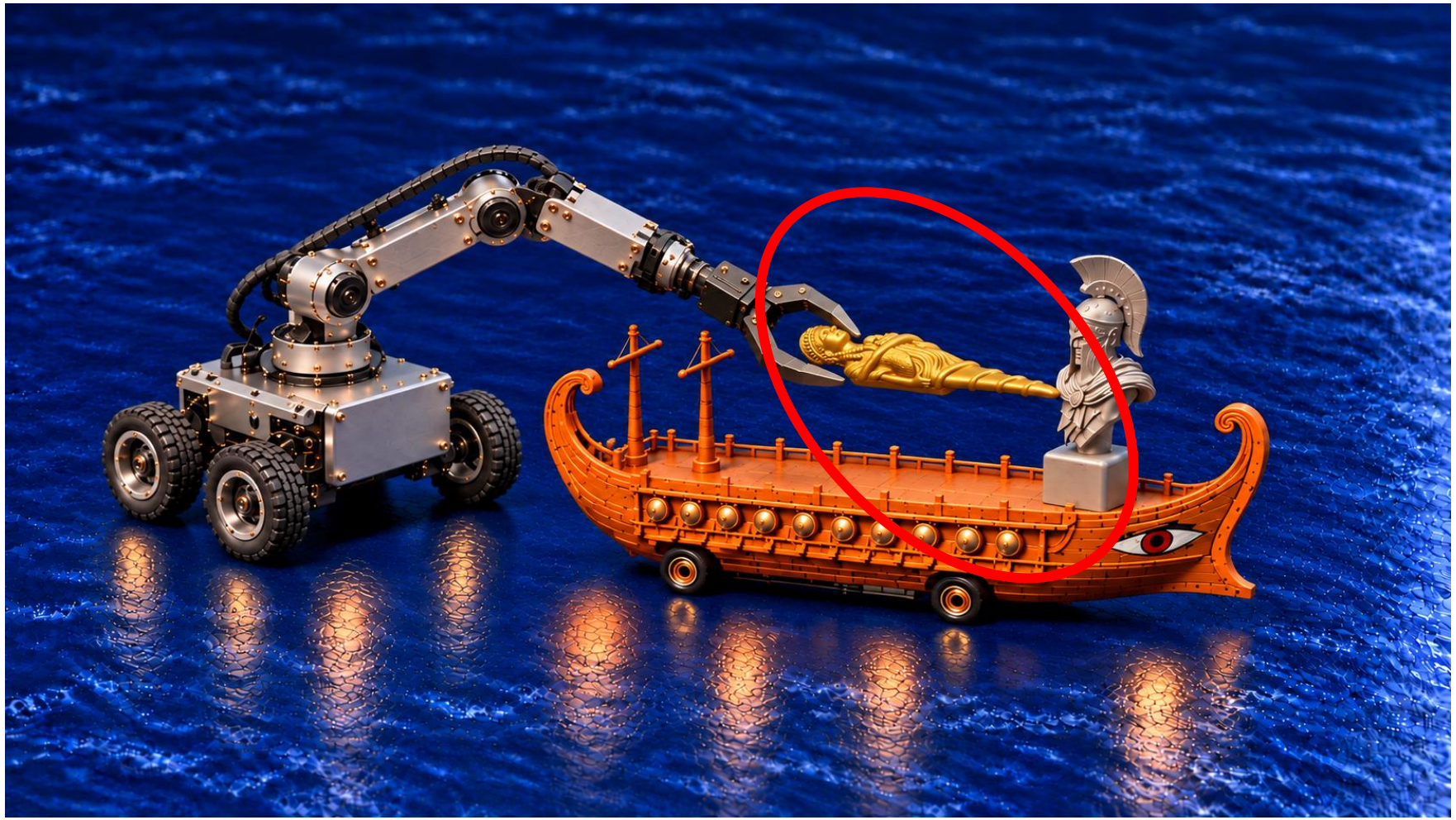
8. The Manual Robot will retrieve Athena Pin from the ocean, then pass to Odysseus

*The pin is located in the ocean somewhere in the map. The location may change randomly.



Game Flow

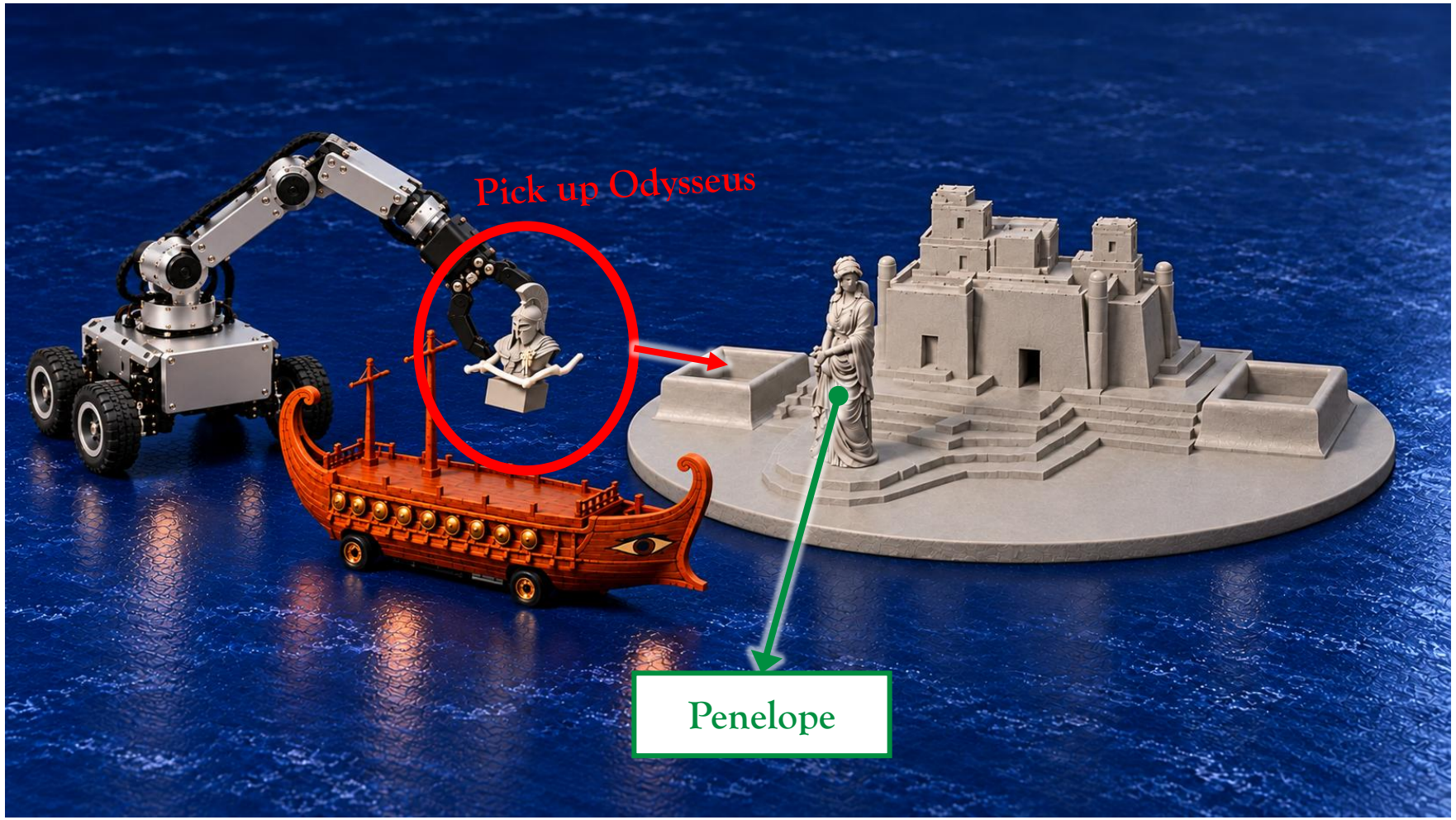
8. The Manual Robot will retrieve Athena Pin from the ocean, then pass to Odysseus
   *The pin is located in the ocean somewhere in the map. The location may change randomly.



Game Flow

9. Finally, the Manual Robot will move Odysseus to Ithaca

*The Ithaca consists of the castle, Penelope, and the area to place Odysseus, the Manual Robot will place Odysseus in the indicated area below. Make sure Odysseus has both Pin and Bow prior to entering Ithaca



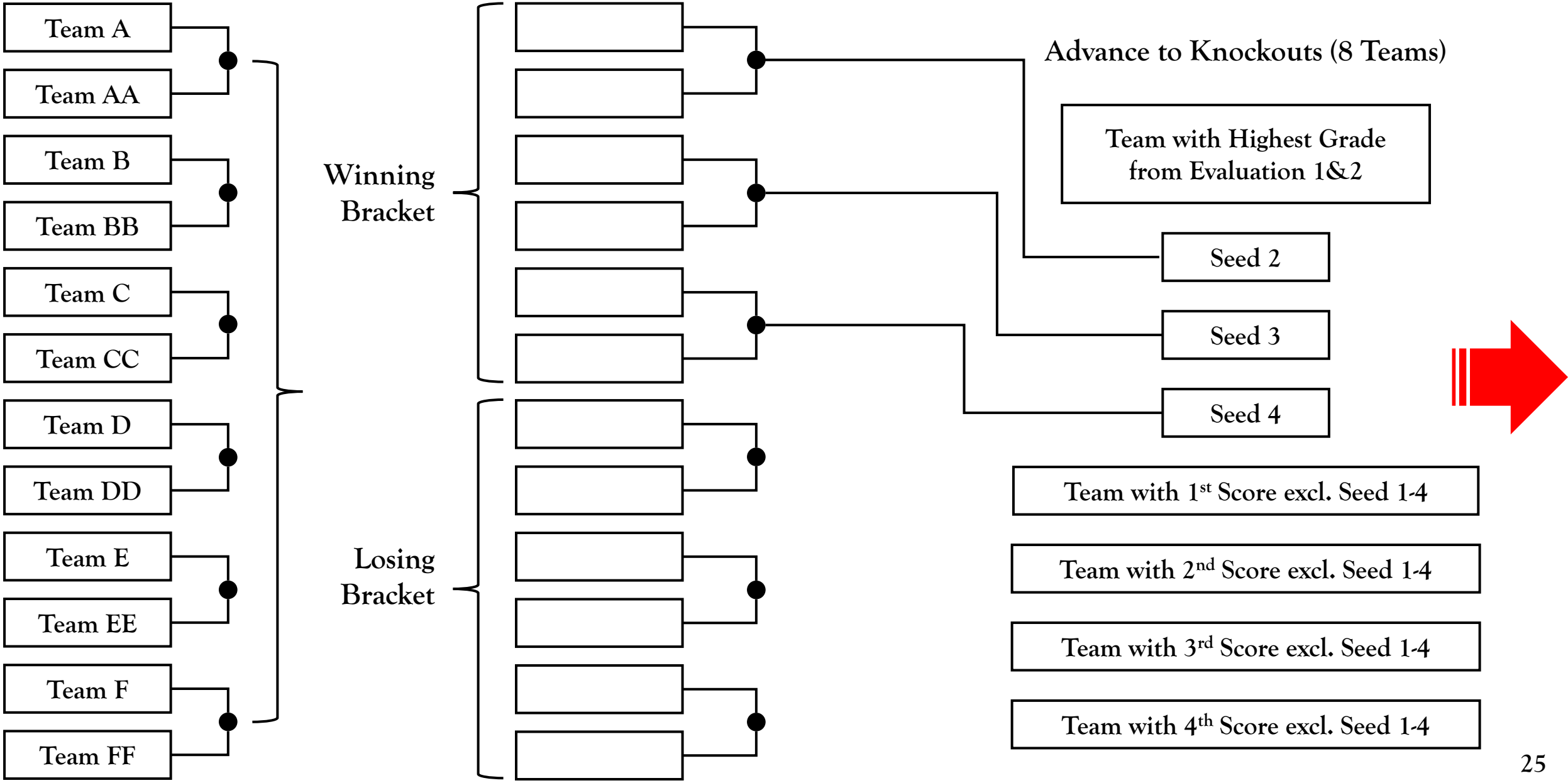
Hidden Mission

10. Hidden Mission can only be access before Odysseus enters Ithaca. Once Odysseus enters Ithaca, the students can no longer earn points. The hidden mission is inside the Darkroom on the playing field. Different images will be displayed on the screen along with the order of trials, and students will need to push side of each cube to rotate them to complete the image.
    *Darkroom is not visible from outside; students must rely on the **camera view** attached on their manual robot
    ~~*The robots can enter the Darkroom after 3:30 time mark~~



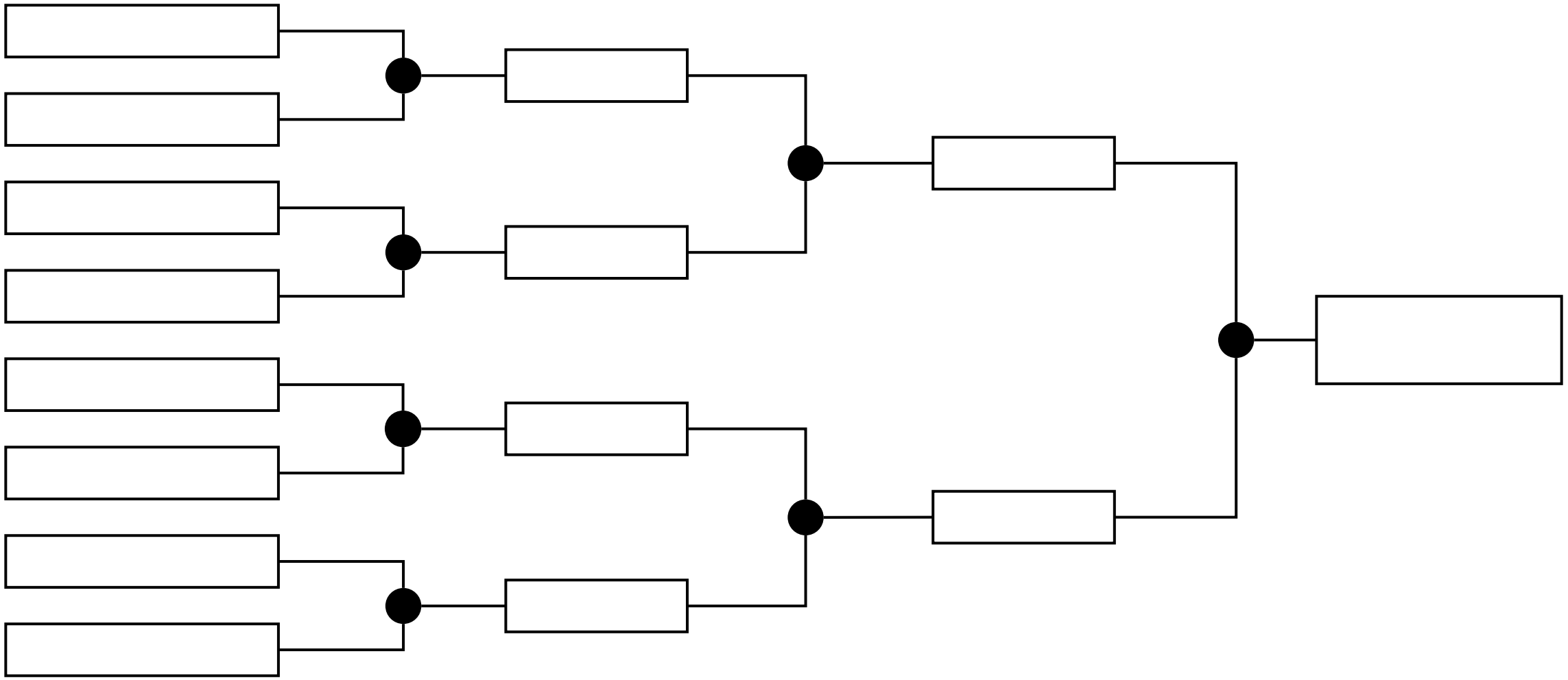
Game Rules

Tournament Schedule (Play In)



Tournament Schedule (Knockouts)

Randomly Distributed



Rules – Summary

Prioritized winning strategies: Returning Odysseus to Ithaca > Total Score > Time of Arrival

Robot Type	Rule	Score
Auto Robot	Single lap following the line (Before Odysseus arrives at Ithaca)	+5/lap
	Successful stop at the statue	+5n
Manual Robot	Successful transfer of crews from Auto Robot to Deathland of Hades	+2x
	Successful transfer of statues from playing field to the altar	+2n
	Successful transfer of Odysseus from Auto Robot to Ithaca +10 for Odysseus, +5 for the Bow, +5 for the Athena Pin	+10/15/20
	Hidden mission (+5 for each cube)	+5/cube
Both	Collision with another team's robot or entering opponent's playing field	-20
	Robot reset	-5

*n = sequence of the trial, kept at 1 for wrong sequence; x = number of crews

General Rules – Details

*The rules may CHANGE based on students' feedback during Evaluations

1. How to Win?
   - If **ONLY one team** successfully sends Odysseus back to Ithaca, that team **win** regardless of the score
   - If **both teams** successfully sends Odysseus back to Ithaca, the team with **higher score** wins
   - If **both teams** fails to sends Odysseus back to Ithaca, the team with **higher score** wins
   - If **both teams** have same score, the team that completes all missions including hidden mission wins
   - If **both teams** have same score, and completed all missions, the team who **completes earlier** wins
2. Time Limit
   - 5 minutes to complete all trials within time including the hidden mission
   - The game **will not end** even if one team completes all missions before 5 minutes mark
3. Physical Interaction with Robot
   - The dimensions will be measured right before the start of the round
   - Once the dimensions are measured, students are no longer allowed to have physical interaction with robot (Not allowed to modify structure or fix)

General Rules – Details

4. Design Rule
   - There will be no limitation on how many manipulators to be installed on Auto or Manual robot as long as the robots are within dimension limitation
5. Line Rule
   - The Auto Robot is not allowed to leave line track, Auto robot will not be allowed to move outside of the track (line)
6. Trial Rule
   - Manual Robot can only transfer statues to altar **only** if required number of soldiers are moved to Deathland of Hades
   - Auto Robot **must stay** at the same position (after recognition) until the statue is moved off from the field. The Manual Robot must lift the statue before Auto Robot starts moving again
7. Hidden Mission
   - Manual Robot can enter the Darkroom any time during the game
8. Score Guide
   - The score gained from completing trial increase as later trial is being completed
   - If any part of the robot is in other team's area (this includes end effector, wheels, chassis, etc) during collision, it lies under "Crashing into opponent's robot" rule

General Rules – Details

9. Additional Material Purchase:
   Each team may use any material worth up to **20,000 ₩**. Students are allowed to purchase transmission components (such as gears, wires, axles, and chains), sensors, and hardware structures. However, controllers, actuators, communication devices, energy storage devices, and energy sources cannot be purchased.

   For any items not specified in the list above, please contact TAs via mail or Classum. Make sure to keep receipts for purchased parts for future verification (the item and price must be shown). The expenses must be included in the **Robot Project Evidential Documents**.

General Rules – Details

10. 3D Printing & Laser Cutting:
    Each team may use either 3D printer or laser cutter to fabricate complicated parts that may be hard to fabricate with the provided tools and materials. The usage limitation for 3D printed parts is as follows:
    - There is no limit on the amount of 3D printing filaments used for fabricating the teleoperated manipulators.
    - However, for the chassis of the robot (Auto & Manual), the filaments usage must not exceed 60g
    - The filaments usage must be tracked and submitted with **Robot Project Evidential Document**
    - There will be charge of **(500 ₩/10g)** which will be included as a part of **20,000 ₩** expense limit, if the filaments usage excess the limit of 60g
        - The filaments usage will be checked prior to ROBOCON
        - Only FDM printers at the Idea Factory & Design Studio along with provided filaments are allowed for usage. The printed parts with other materials (such as TPU, metal, etc) or other printers are not allowed

Score Guide – Details (Difficult version)

+5/lap If **Auto Robot** runs a lap following the line, this will be counted if the robot travels back to initial position in the same orientation. For example, if the robot follows the guideline and runs a full lap, it is counted as one lap; However, if the robot goes forward and moves backward to reach initial position, it is NOT counted as a lap. The **maximum of 5 x (5 + number of trials)** points can be earned through this scoring method.

+5n If **Auto Robot** successfully stops by the statue after detection, the team will receive **5n** points based on the sequence of the trials. The "n" is sequence order of the trial. If the robot stops at the incorrect position, only 5 points will be granted. For example, if the trial type A is displayed as third trial and the team's **Auto Robot** successfully stops by the type A statue, 5 x 3 = 15 points will be provided. How to determine if the **Auto Robot** stopped at a correct position? There will be a line connecting the statue and the line track (normal to the line) and if and only if part of the robot is stepping on **the mark**, it will be considered success.

+2x If **Manual Robot** successfully transfer required number of crews from **Auto Robot** to **Deathland of Hades** per trial, the team will receive **2x** points based on the sequence of the trials. For example, if the third trial type A requires 2 crews to be transferred, team can only receive 2 x 2 = 4 points if 2 crews are properly transferred to **Deathland of Hades**. The team MUST make sure that the crews do not fall out from the **Deathland**.

Score Guide – Details

+2n If **Manual Robot** successfully transfer trial statue from the playing field to the altar per trial the team will receive **2n** points based on the sequence of the trials. For example, after successfully moving required number of crews to the **Deathland of Hades**, **Manual Robot** can transfer the statue to the altar to receive points. The statues do not need to be placed in specific location per trial, as long as the statue fits inside the hole in the altar, it is considered success

+10(40) If **Manual Robot** successfully transfer Odysseus from auto robot to Ithaca. Odysseus must have **Athena Pin** and the **Bow** with him **prior to** entering Ithaca. Once Odysseus enters Ithaca, the team can no longer earn points afterwards. Each Bow and Athena Pin is considered +5 points each. The team who transfer Odysseus to Ithaca first receive double the point. For example, if Odysseus enters Ithaca (+10) only has Bow (+5) with him but arrived first, the team will receive 2 x (10 + 5) = 30 points.

+5(45) If **Manual Robot** successfully completes the hidden mission prior to entering Ithaca, for each puzzle completed, the team will receive +5 points each. For example, if the hidden mission is designed to have 3x3 puzzle, and the team completes 7 out of 9 puzzles, they will receive +35 points.

Score Guide – Details (Penalty)

-20 If **Any Robot** enters opponent's playing field area or collides with opponent's robot, they will lose 20 points. For example, if the **Manual Robot** falls into opponent's playing field area from upper floor, the Manual Robot's team will lose 20 points.

-5 If **Any Robot** needs to be reset. For example, if the team needs to reset robot, the timer will NOT stop and they MUST place robots in the starting position, then lose 5 points.

Especially if robot enters opponent's playing field area, the robot will be **forced reset** and moved back to starting position. This implies that the team will lose total of 25 points from entering opponent's field (-20) and reset (-5)

Robot Constraints

Auto Robot

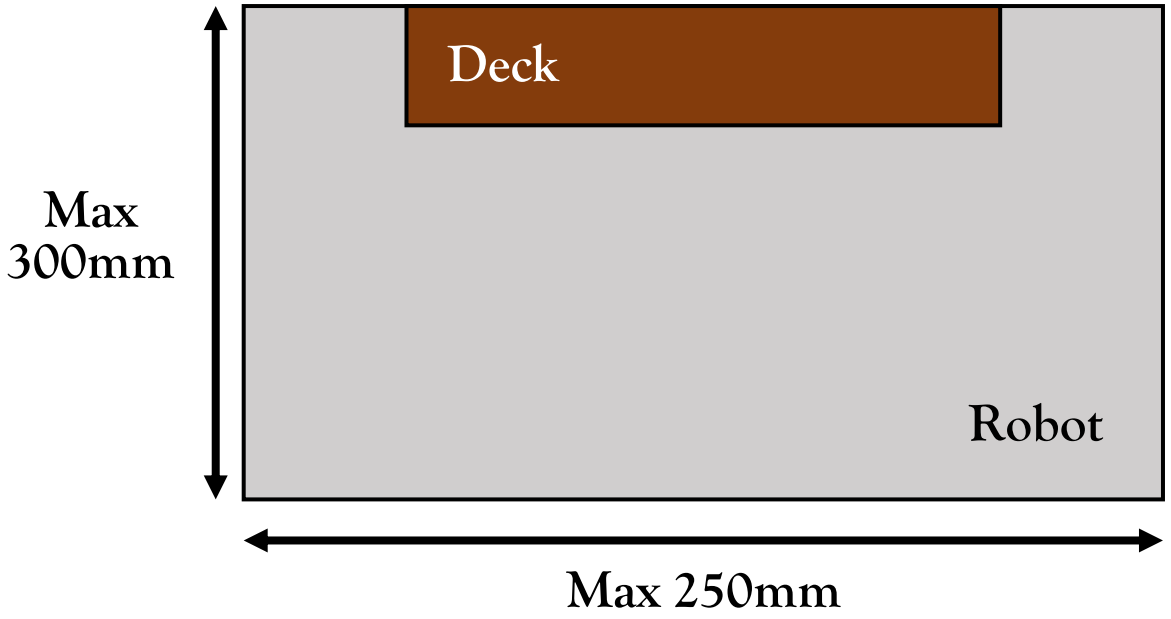


Size Restriction

260mm x 260mm x 300mm / 4.0kg

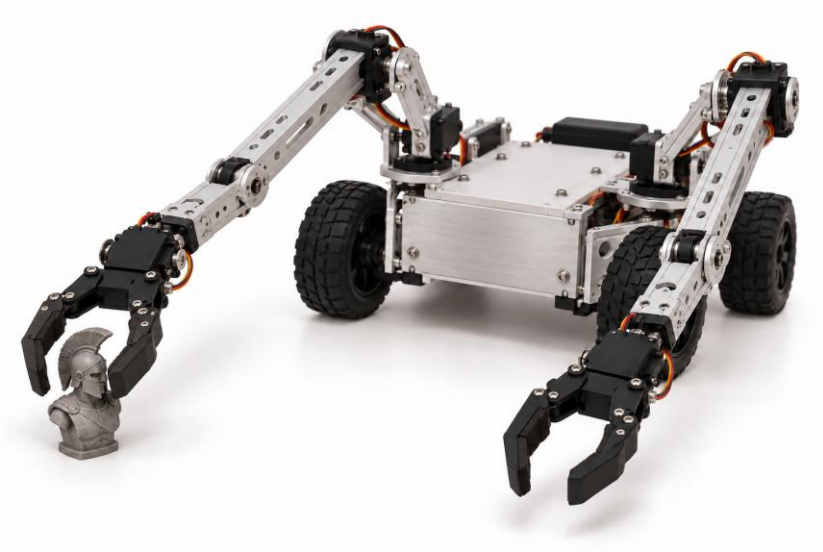
Needs **clearance area** for the deck

Deck size: 200mm x 120mm x 15mm / 350g



Robot Constraints

Manual Robot



Size Restriction

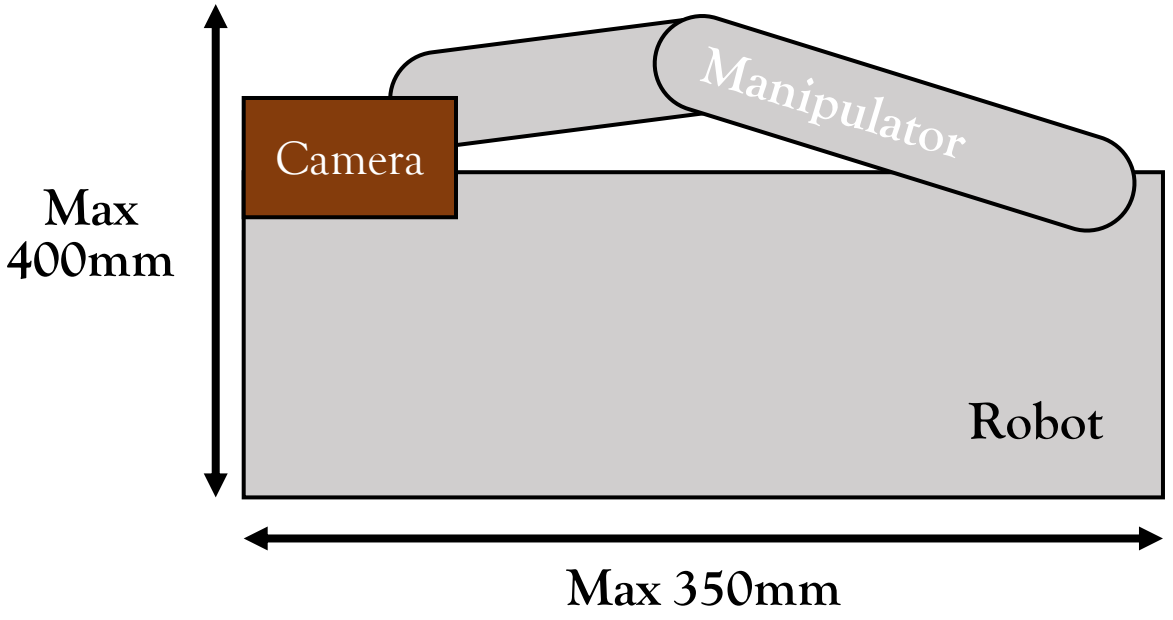
350mm x 350mm x 400mm / 6.0kg

No restriction on number of manipulators

Needs **clearance area** for the camera

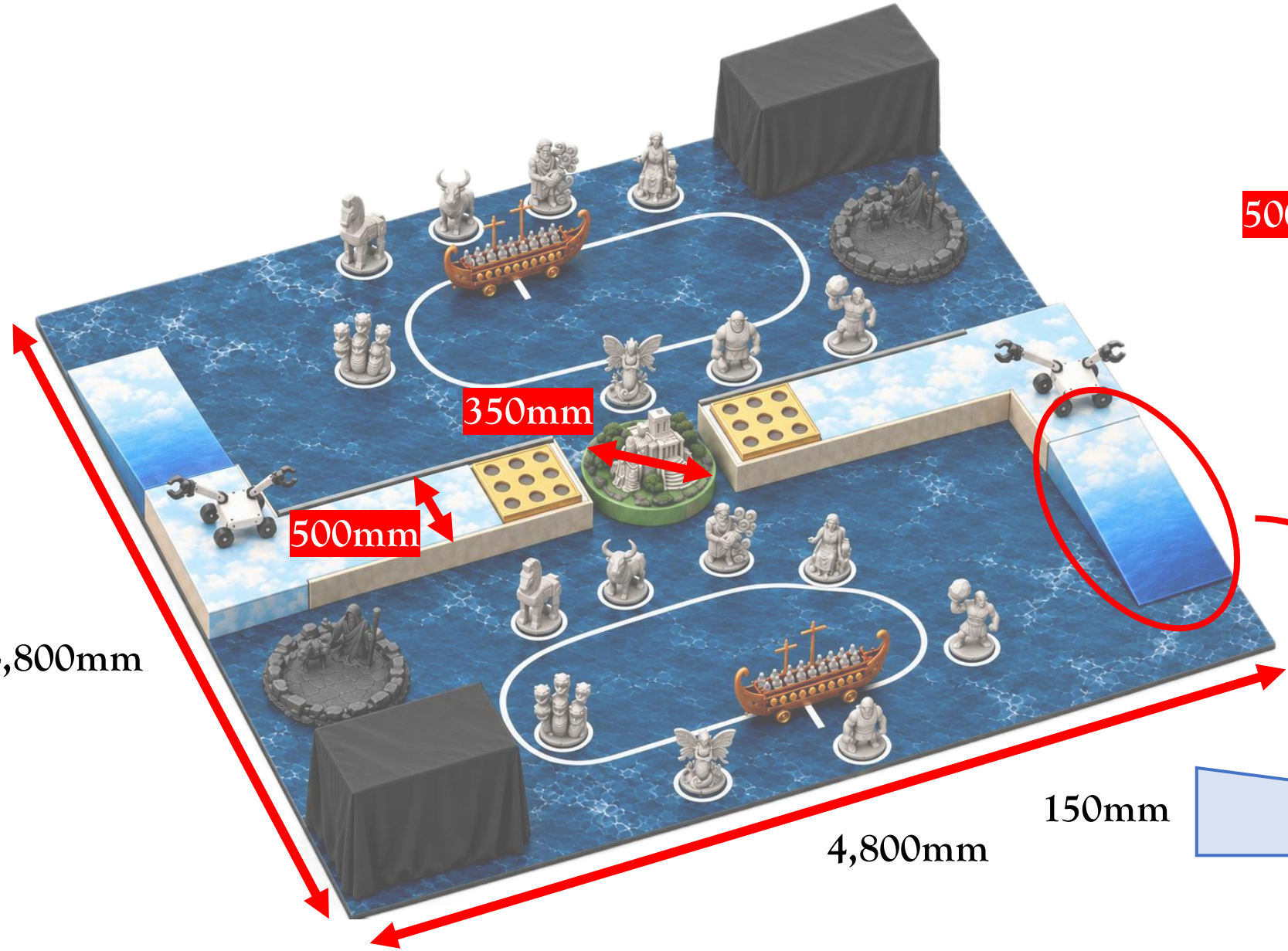
Camera size: 70.5 x 44.2 x 32.8mm / 145g

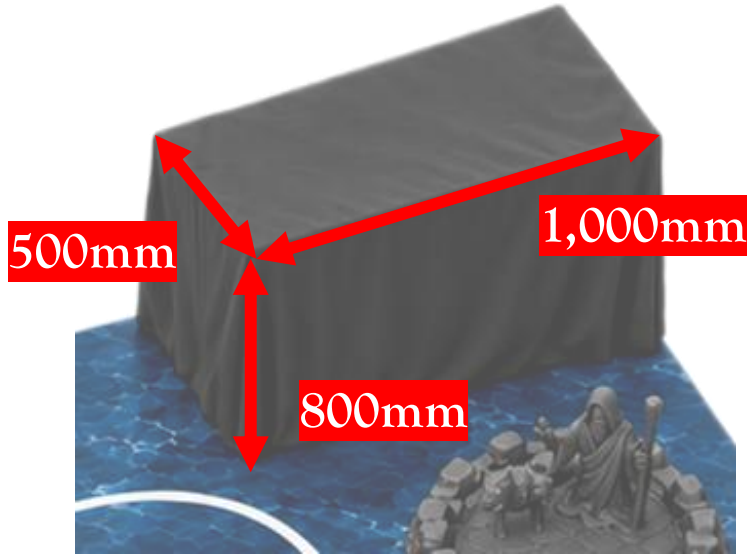
Folded Manipulator



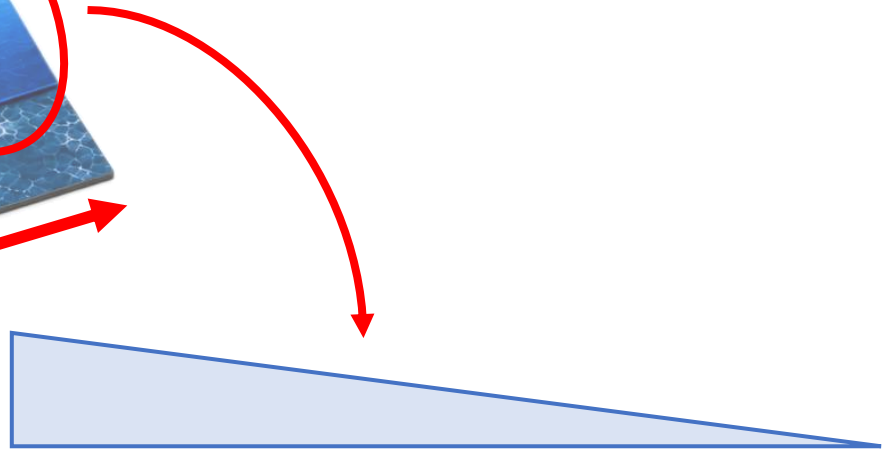
Supplementary

Map Dimension





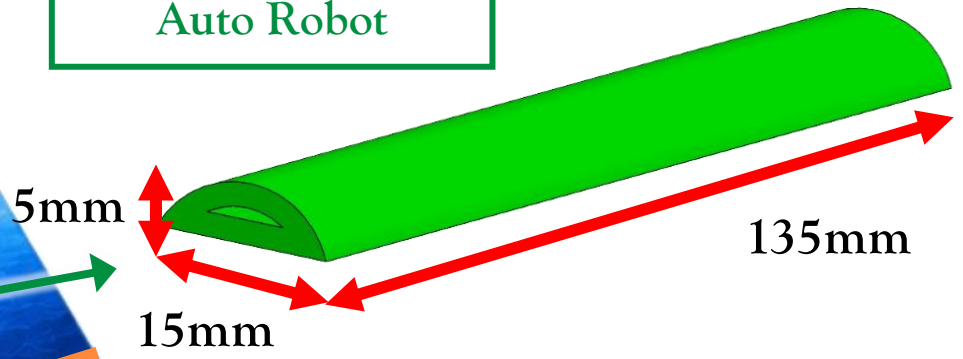
Darkroom



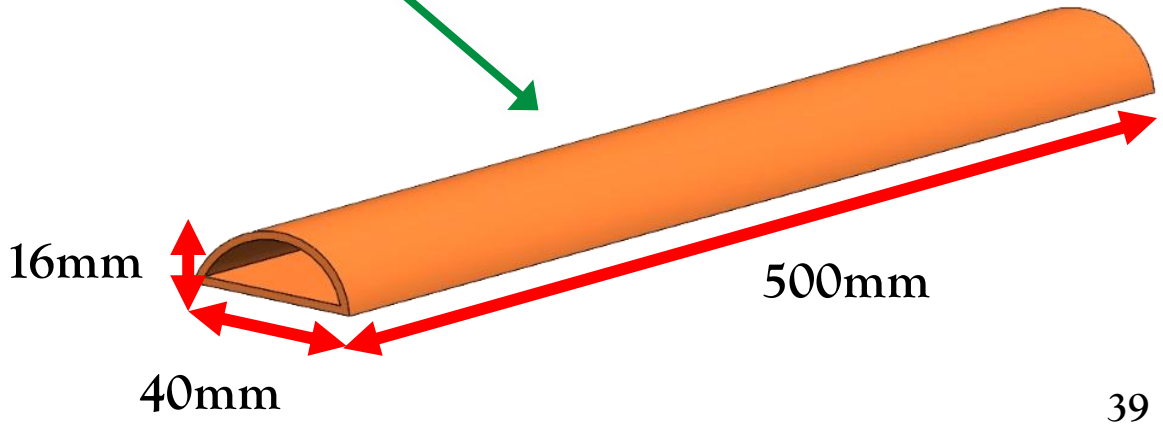
Map Dimension



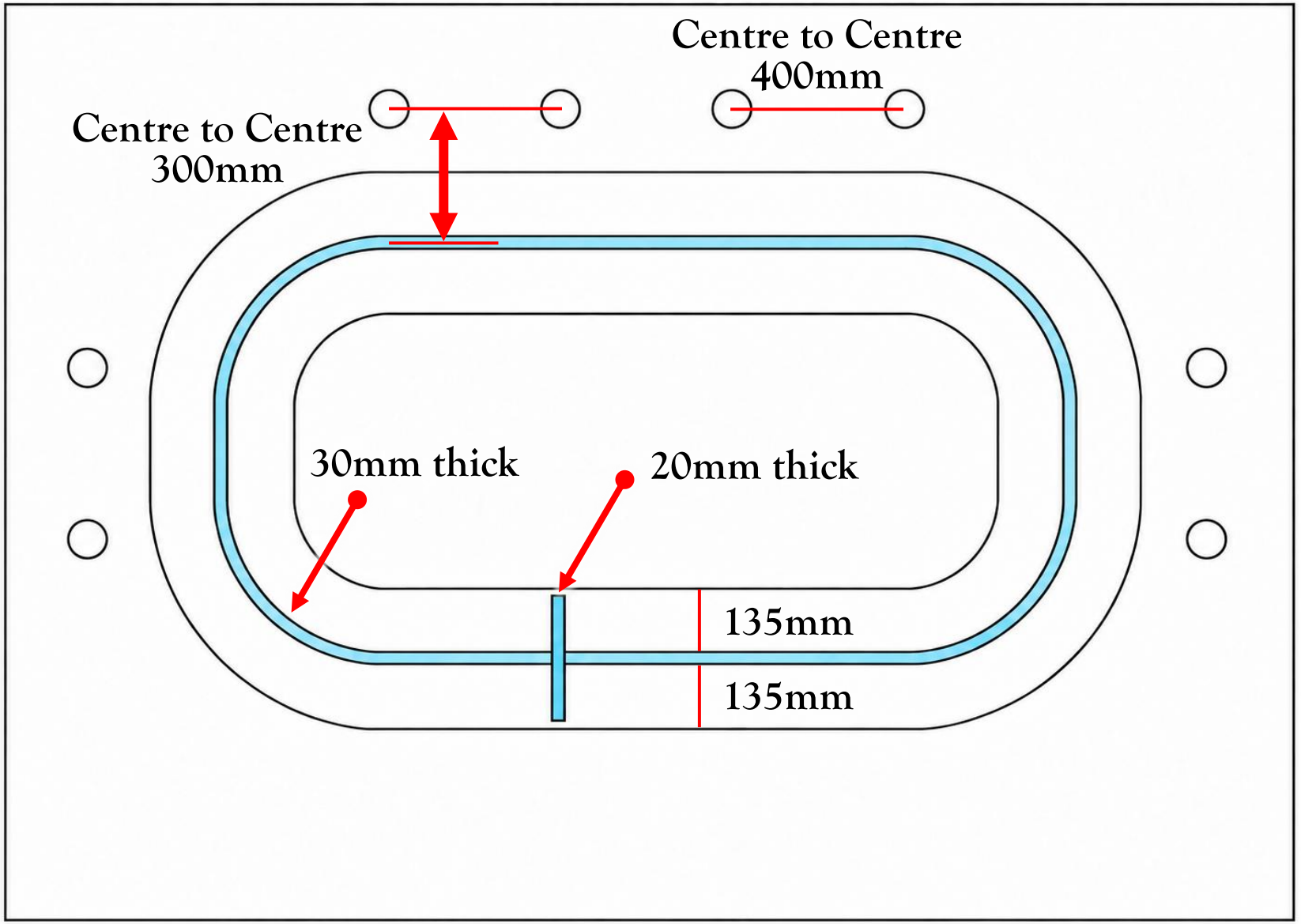
Speed bump for Auto Robot



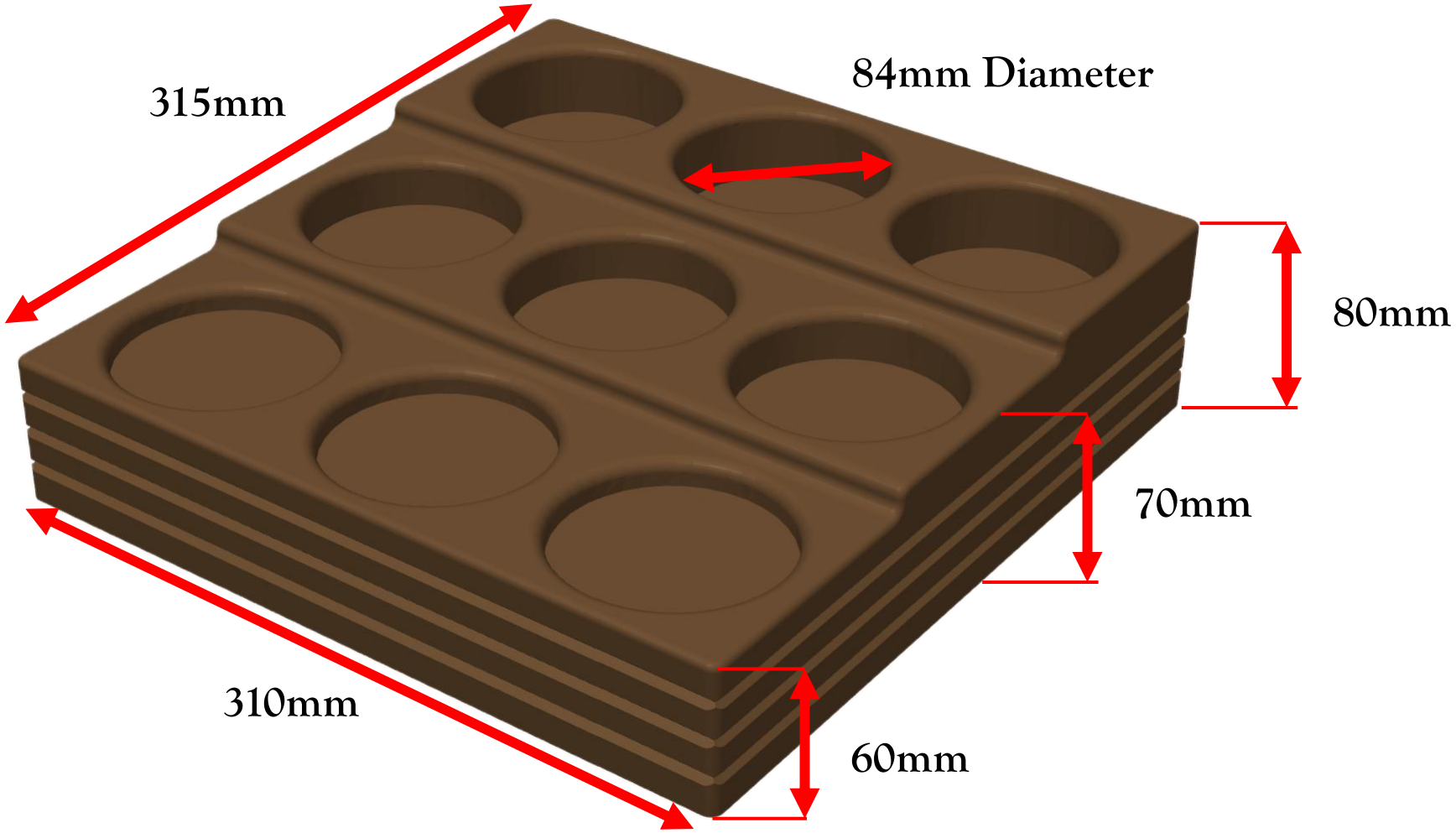
Speed bump for Manual Robot



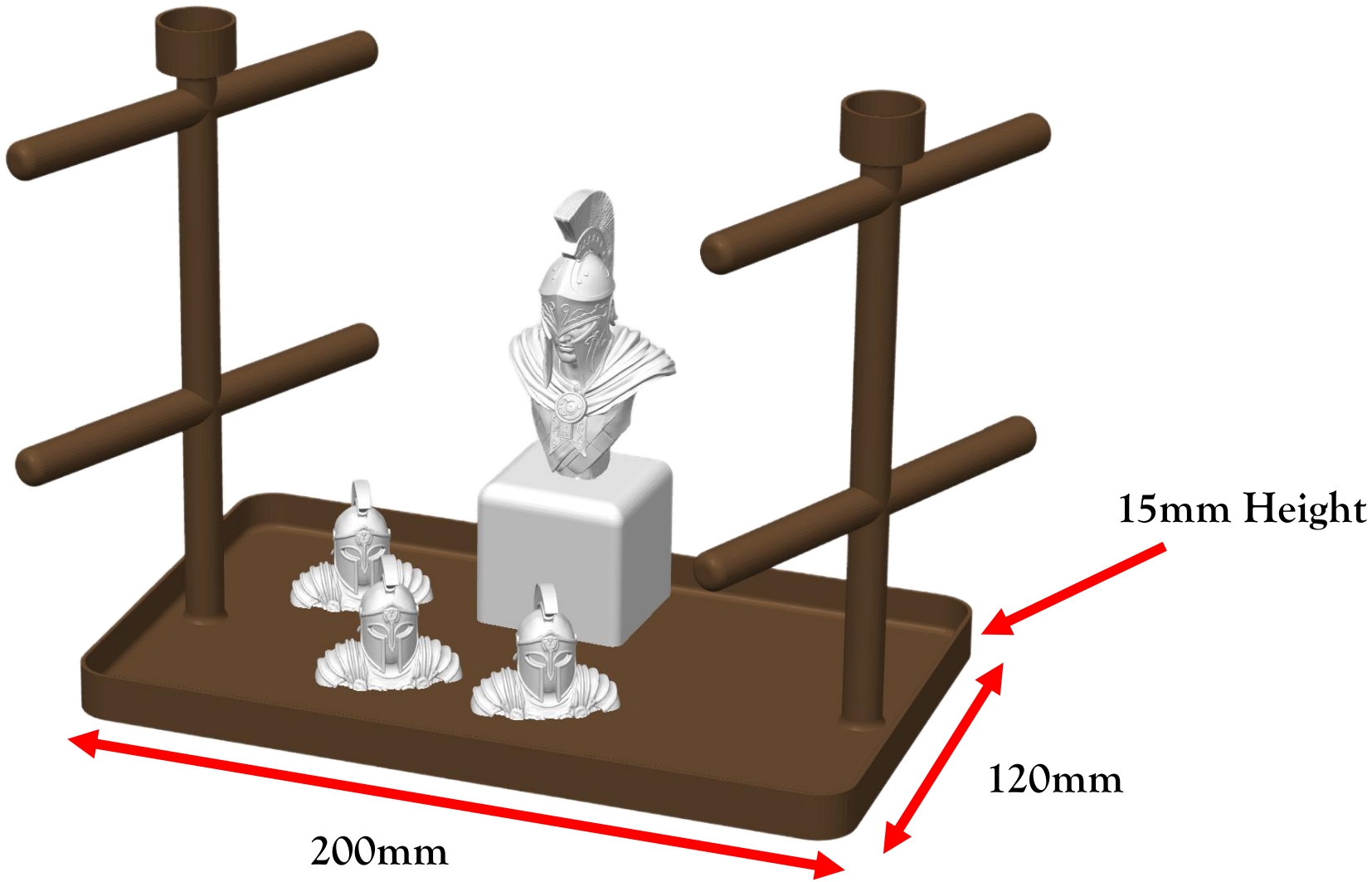
Map Dimension



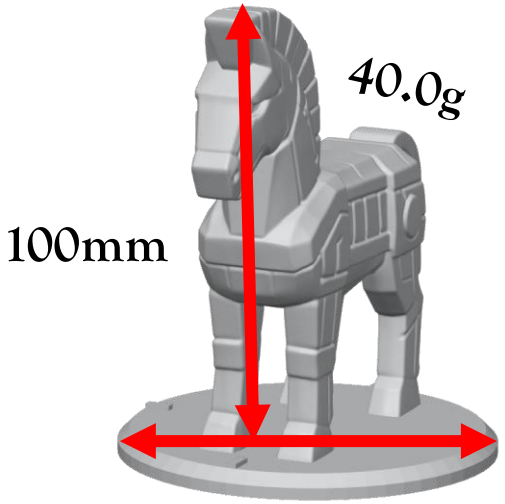
Altar Dimension

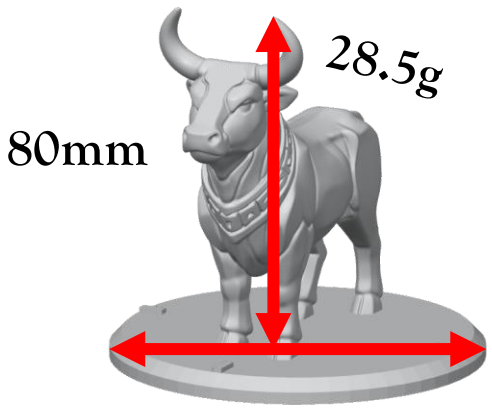


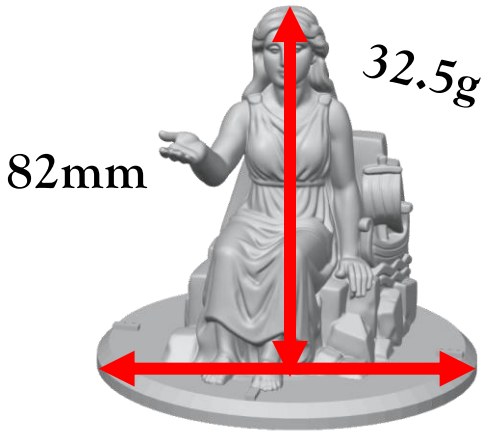
Deck Dimension

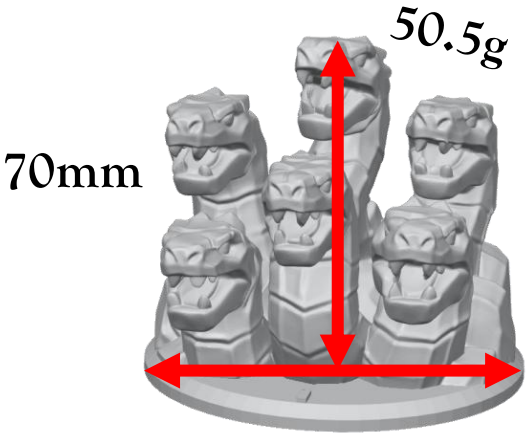


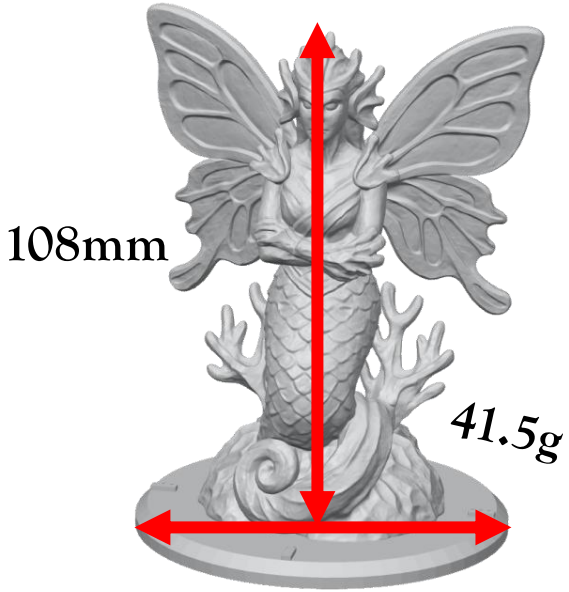
Statues Dimension

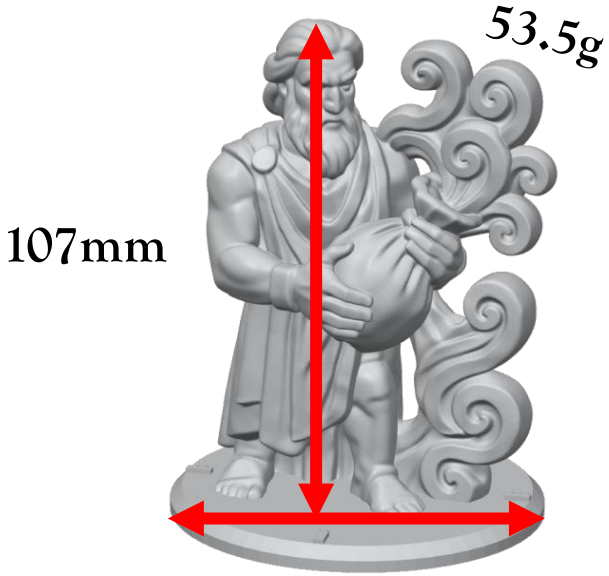


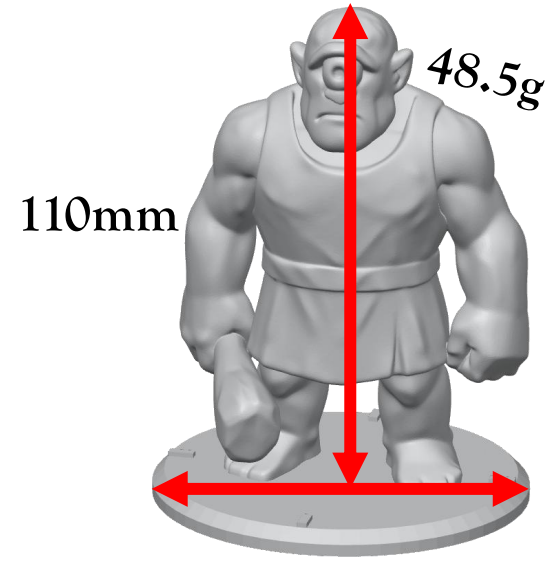


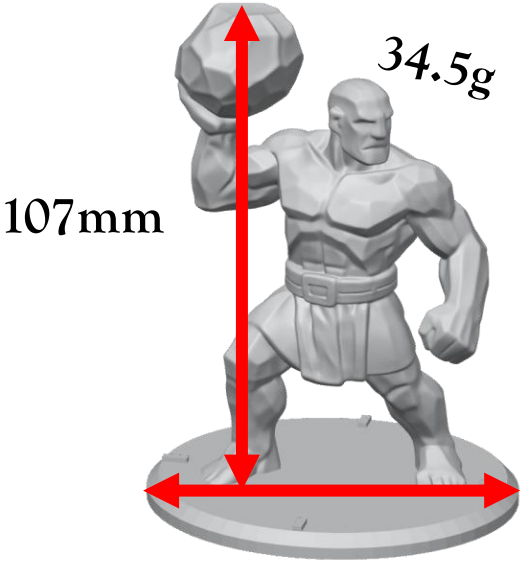












Other Important Objects Dimension

